

Paralog Compensation-Based Synthetic Lethality Discovery

DepMap 26Q1
1,208 cell lines

HGNC Gene Families
66,595 paralog pairs

CPTAC Proteomics
232 HGSOC samples

TCGA + PRISM
Survival + Drug

Delta Dependency (DD)

$$DD = \text{mean}(\text{CERESparalog} \mid \text{drivermutant}) - \text{mean}(\text{CERESparalog} \mid \text{driverwildtype})$$

+ Sequence Identity Filter ($\geq 30\%$)

+ Normal Cell Toxicity Filter + PRISM Drug + CPTAC Protein

De Novo SL
Candidates

Cross-Cancer
Validation

Clinical
Translation